

Muthuraja S

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PROFESSIONAL SUMMARY

Computer Science student with strong foundations in Data Structures, Algorithms, and systems design. Experienced in building production-grade backend services, containerized applications, and cloud deployments with a proven problem-solving track record (LeetCode Knight, rating 1852).

EDUCATION

Indian Institute of Information Technology, Sri City, Tirupati, AP

Aug 2023 – May 2027

- B.Tech in Computer Science and Engineering (**CGPA: 7.9/10.0**)
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Cloud Computing, C/C++

EXPERIENCE

Backend Development Intern, ResoluteAI Software

May 2026 – Aug 2026

- Engineered production-grade, object-oriented RESTful APIs using FastAPI, implementing structured request lifecycles, validation models, authentication, and centralized error handling.
- Designed and indexed scalable MongoDB schemas, optimizing query execution plans and access patterns for high-throughput data operations.
- Authored modular, maintainable code following industry standards and OpenAPI/Swagger documentation, collaborating across teams to streamline backend service integrations.

PROJECTS

AgentMesh | [GitHub](#) | agentmesh.muthuraja.dev

- Architected a multi-tenant B2B decision platform using an OOP multi-agent model (SAGE/GUARDIAN/ORACLE/EMPATH) with deterministic fail-closed state machines and immutable hash-chained audit logs. Designed PostgreSQL/pgvector relational schemas with tenant isolation, asynchronous Celery task queues, and hybrid RAG retrieval. **Tech Stack:** FastAPI, React, TypeScript, PostgreSQL, pgvector, Redis, Celery, Docker, Azure.

Meta-Agent Framework | [GitHub](#) | meta.muthuraja.dev

- Built a modular orchestration engine with a Directed Acyclic Graph (DAG) runtime to dynamically instantiate and execute specialized agents from natural language. Implemented autonomous planning loops, stateful abstractions, parallel task scheduling with Redis, and end-to-end distributed tracing using OpenTelemetry. **Tech Stack:** Python, FastAPI, LangChain, LangGraph, Redis, ChromaDB, Docker, NetworkX.

CargoLink | [GitHub](#) | cargolink.muthuraja.dev

- Developed a scalable logistics marketplace featuring real-time bidding, JWT/OAuth 2.0 RBAC, and WebSocket event-driven state updates. Designed MongoDB data models, integrated Redis caching for low-latency queries, and containerized deployment with automated CI/CD and Jest unit testing. **Tech Stack:** React, Node.js, Express.js, MongoDB, Redis, WebSockets, Docker, GCP, Jest.

SheetSnap | [GitHub](#) | ocr.muthuraja.dev

- Created an AI document digitization tool for tabular extraction; implemented client-side parsing pipelines, multithreaded PDF handling, structured state management with Zustand, and automated unit testing via Vitest. **Tech Stack:** Next.js, React, TypeScript, Gemini AI API, Zustand, pdfjs-dist, Vitest.

TECHNICAL SKILLS

Languages: Java, C, C++, Python, JavaScript, TypeScript, SQL, HTML/CSS

Core: DSA, Object-Oriented Design, Data Modeling, RESTful APIs, Microservices

Backend, Databases & Distributed: Node.js, Express.js, FastAPI, Celery, WebSockets, Kafka, Redis, PostgreSQL, MongoDB, MySQL, ChromaDB, pgvector

Frontend & Frameworks: React.js, Next.js, Vite, Tailwind CSS, Zustand

DevOps, Cloud & Testing: Linux, Bash, Git, Docker, GitHub Actions, OpenTelemetry, GCP, Azure, Jest, Vitest, CI/CD

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

PEC Hacks 4.0 – 1st Place | Panimalar Engineering College, Chennai

Aug 2026

- Won first place (₹1 Lakh cash prize) at PEC Hacks 4.0 (36-hour MLH-backed international hackathon) by building FinGuard – a pre-payment safety layer for UPI QR codes featuring deterministic risk scoring, real-time QR parsing, and explainable fraud detection. [\[Certificate\]](#)

LeetCode Knight | leetcode.com

2025

- Contest rating of **1,852** (Knight badge), demonstrating strong problem-solving proficiency in Data Structures, Algorithms, and space-time complexity analysis.

Smart India Hackathon (SIH), Campus Level Finalist

2025

- Selected at the campus level for SIH 2025, competing among top teams to solve real-world problem statements set by government ministries and industries.

Web Development Domain Lead, Nexsync (College Club)

09/2024 – Present

- Leading the Web Dev team, steering architectural decisions, and conducting technical workshops and mentoring sessions for 100+ students.